AI 101 for Mechanical Engineers · Module 9

Grouping, compression and fair selection

Lesson 9.3

What you will learn

> Describe k-means and principal component analysis, explain the effect of feature scaling, and apply a fair protocol for comparing candidate models.

Lesson overview

Module 9 · Lesson 3

Grouping, compression and fair selection

AI 101 for Mechanical Engineers

1 k-means groups unlabelled data



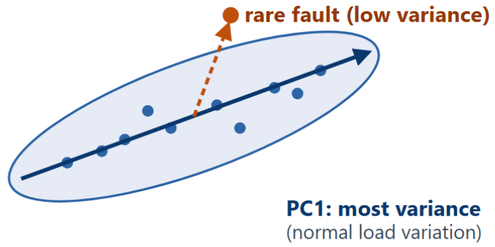
× = cluster centre
k = 3 chosen
by the user

Groups operating profiles around centres by minimizing within-cluster distance.

Choosing k: the elbow method compares within-cluster variation across several k. It is an aid, not an objective answer; check stability and meaning.

k is not the number of physical faults or machine states.

2 PCA compresses by variance



Projects data onto directions of high variance, ranked by the variance they capture.

High variance is not the same as fault relevance. A rare fault may lie in a low-variance direction.

3 Compare candidates fairly

1. **Common development data**
   Every candidate uses the same examples.
2. **Training folds**
   Fit each model and its preprocessing here only.
3. **Validation folds**
   Tune hyperparameters and choose a candidate.
4. **Locked final test**
   Evaluate the fixed choice once.

Apply each training fold's stored transformations to its validation fold without refitting. Report the split, selection rule and test result.

Scaling matters An unscaled large-magnitude variable can dominate distances and make a small but relevant vibration feature negligible. Record the units and scaling used.

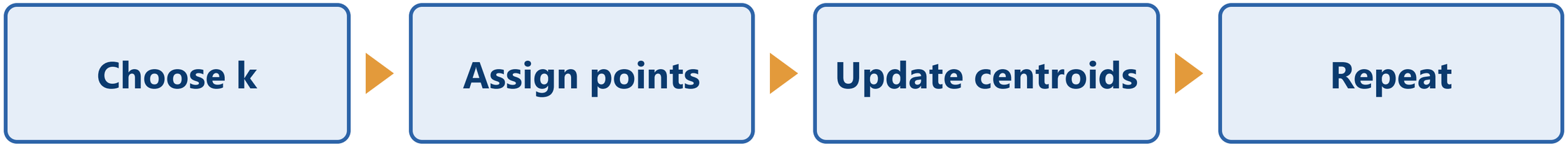
Key takeaway

Clusters and components are mathematical summaries, not fault states.
Compare candidates under the same information boundaries, and justify added complexity with evidence.

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k-means alternates assignment and centroid updates



- Reduces the within-cluster sum of squared Euclidean distances [91]
- Usual algorithm follows Lloyd’s quantization method [51]
- May stop at a local minimum; often repeated from several starts [91]

A cluster count is not a fault count

- k is a user-selected algorithmic setting
- It does not prove the number of physical faults
- Elbow method: look for diminishing improvement as k increases
- The elbow is not guaranteed clear or objectively correct

> Check cluster stability and physical meaning; grouping still needs engineering interpretation.

High variance is not necessarily task relevance

PCA

- Projects onto high-variance directions [52]
- Can compress correlated sensor features
- Ranks components by captured variance

Risk

- Normal ambient-load variation may dominate
- A rare fault may lie in a low-variance direction
- Components are not filtered by fault relevance

> Capturing variation and identifying diagnostically useful information are different questions.

Unscaled large variables can dominate distances

- Distance-based methods need scaling that respects physical units
- A large unscaled variable can dominate distance calculations
- A smaller, task-relevant vibration feature can become negligible
- Choose preprocessing for the task; record the units used

Fair comparison keeps information boundaries



- Compare candidates on common development data [53]
- Fit preprocessing inside each training fold; tune on validation
- Same boundary for every candidate; report split and selection rule

Complexity must justify its cost

- Consider output, data structure, performance evidence and operational cost
- Complex architectures must show gains justifying added maintenance burden
- Variance, clusters or boundaries alone do not establish faults
- KMeans and PCA [91]: mathematical summaries that do not establish fault states

CHECK YOUR UNDERSTANDING

Could the direction of greatest variance be unrelated to a rare fault? Explain.

Answer before moving on.

CHECK YOUR UNDERSTANDING

Could the direction of greatest variance be unrelated to a rare fault? Explain.

> Yes. PCA maximizes mathematical variance. A rare fault can occupy a low-variance direction, so may not align with the greatest-variance component.

CHECK YOUR UNDERSTANDING

Does choosing a cluster count establish the number of mechanical fault types?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does choosing a cluster count establish the number of mechanical fault types?

> No. The count is an algorithmic setting. Grouping still requires engineering interpretation and does not guarantee distinct physical fault classes.

CHECK YOUR UNDERSTANDING

Where should preprocessing be fitted when comparing candidates across folds?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Where should preprocessing be fitted when comparing candidates across folds?

> Inside each training fold. Validation supports tuning and selection; an isolated locked test evaluates the final choice.

Key takeaways

- k-means reduces within-cluster squared distances; k is not a proven fault count
- PCA captures high-variance directions, which may not be task-relevant
- Fit preprocessing in training folds; tune on validation; lock the test
- Justify complexity with task-relevant validation and operational cost

References

[51] S. P. Lloyd, “Least squares quantization in PCM,” IEEE Trans. Inf. Theory, vol. 28, no. 2, pp. 129–137, 1982. doi: 10.1109/TIT.1982.1056489

[52] I. T. Jolliffe and J. Cadima, “Principal component analysis: A review and recent developments,” Philos. Trans. R. Soc. A, vol. 374, no. 2065, Art. no. 20150202, 2016. doi: 10.1098/rsta.2015.0202

[53] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” J. Mach. Learn. Res., vol. 11, pp. 2079–2107, 2010.

[91] scikit-learn developers, scikit-learn User Guide and API Reference, ver. 1.9.1. [Online]. Available: https://scikit-learn.org/stable/ Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.